Wires, Wi-Fi and Protocols

Data Transfer Media

All networks must enable computers to communicate with one another. This can be done using two main data transfer mediums:

1. Cables (wired)
2. Wi-Fi (wireless)

Cables (wired)

There are two main types of cable which can transport data between computers:

Ethernet Cables

These are made up of copper wires with the data being transported via electrical signals. Because they carry electrical signals, they can suffer from interference.

Don't get the term 'Ethernet Cable' and the word 'Ethernet' muddled up. 'Ethernet' is a protocol (a rule) governing how data should be transported, whereas 'Ethernet Cables' are the media through which the data is transported.

Fibre Optic Cables

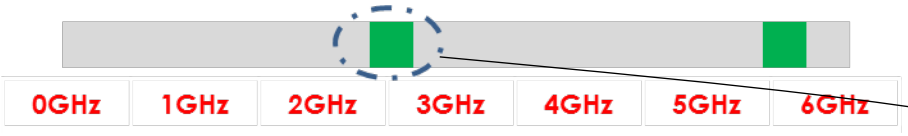
These are made up of glass fibres with the data being transported via light signals. Fibre optic cables are more effective...nothing travels faster than the speed of light! They also do not suffer from interference. They are however more expensive than Ethernet cables.

Wi-Fi (wireless)

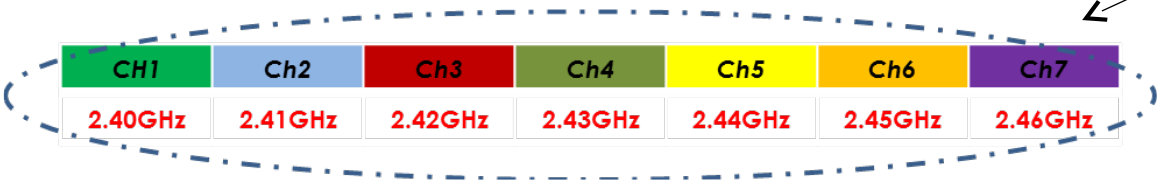
Wi-Fi allows network access to portable devices by using radio waves to send data (not wires!). Wi-Fi is a 'standard'. This means that all Wi-Fi devices are using the same standard technology. In particular they send and receive data using standard radio frequency bands:

2.4 GHz Band (frequencies between 2.4GHz & 2.5GHz)

5 GHz Band (frequencies between 5.725GHz & 5.875GHz)



Furthermore, each band is split into channels that cover smaller frequency ranges, within the band.



This is so that if two devices are sending signals using the same band (e.g. 2.4GHz), they can each use different channels and therefore slightly different frequencies (e.g. 2.41GHz and 2.45GHz), so that they do not suffer from interference with one another.

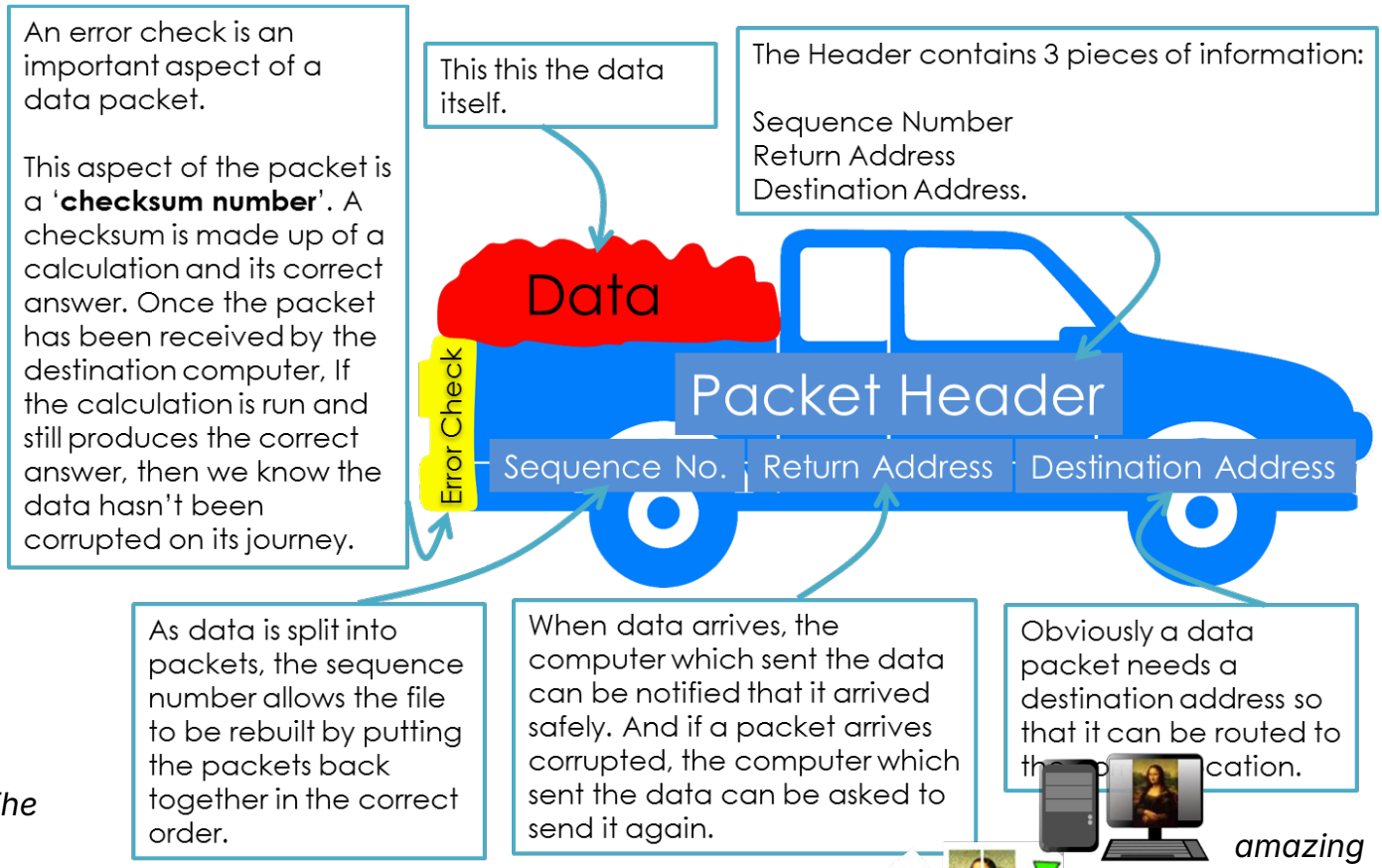
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Data Packets

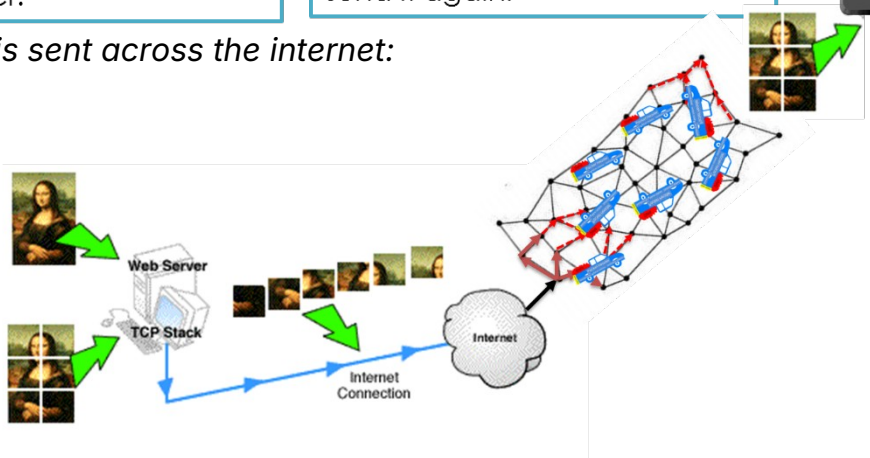
We have already learnt how the Internet is a massive network of networks, but how does data travel around the ginormous network? Data packets are the answer!

When files are sent over the internet they are split into millions of data packets. Packets get sent by different routes according to availability. When you send a file online, the parts of the file might travel one way around the world and the other parts may go in the opposite direction! Packets are reassembled at receiving end.

Typical packet structure:



The amazing way that data is sent across the internet:



IP addressing, MAC addressing, and protocols

So, we have now seen how data is sent around networks in data packets, but because the internet has billions of computers and devices attached to it, these devices will only be able

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to communicate with one another if they 'speak the same language' (i.e. if these data packets are produced and sent in the same standard way). Because of this, the internet is full of protocols, which is really just a fancy name for rules. By manufacturing nodes and network devices to adhere to these rules, all devices on the internet can work effectively. We shall now take a look at some of the major protocols which enables the internet to work. They sound scary but they really are not!

MAC Addresses

Just like humans have postal addresses, every network card in the world has a MAC address (MAC - Media Access Control). A MAC address is hard coded into each and every network device and cannot change. It is a 12-digit hexadecimal number (e.g.

56:b3:8f:90:64:ef). It is there so that data being sent over a network can be sent to the correct network card. It is known as a PHYSICAL ADDRESS

IP Addresses

Just like each network card will have a MAC address, every computer on a network will have an IP address. They are usually a 32bit number written as 4 decimal numbers (e.g.:

129.168.1.101). They are required for the same reason...so that data being sent over a network can be sent to the correct node. The main difference is that an IP address is not hard coded into the computer. It can change and is therefore said to be dynamic. The reason for this is so that on a network, similar types of hardware can be grouped by being assigned similar IP addresses. For example, computers could all be given an IP address starting 190, printers could be given an IP address staring 200 and so on. This means that different types of device can be more easily managed on a network.

The TCP/IP Protocol

This is probably the most important protocol as it determines how all data is sent over the internet. Actually it is made up of two protocols:

TCP – Transmission Control Protocol

We saw earlier how data is split into packets before being sent across the internet. The TCP defines the rules for this.

- It ensures all data is split up into data packets in the same way.
- It ensures that these data packets are put back together when received by the destination device.
- It also ensures that any data received is the same as that sent.

IP – Internet Protocol

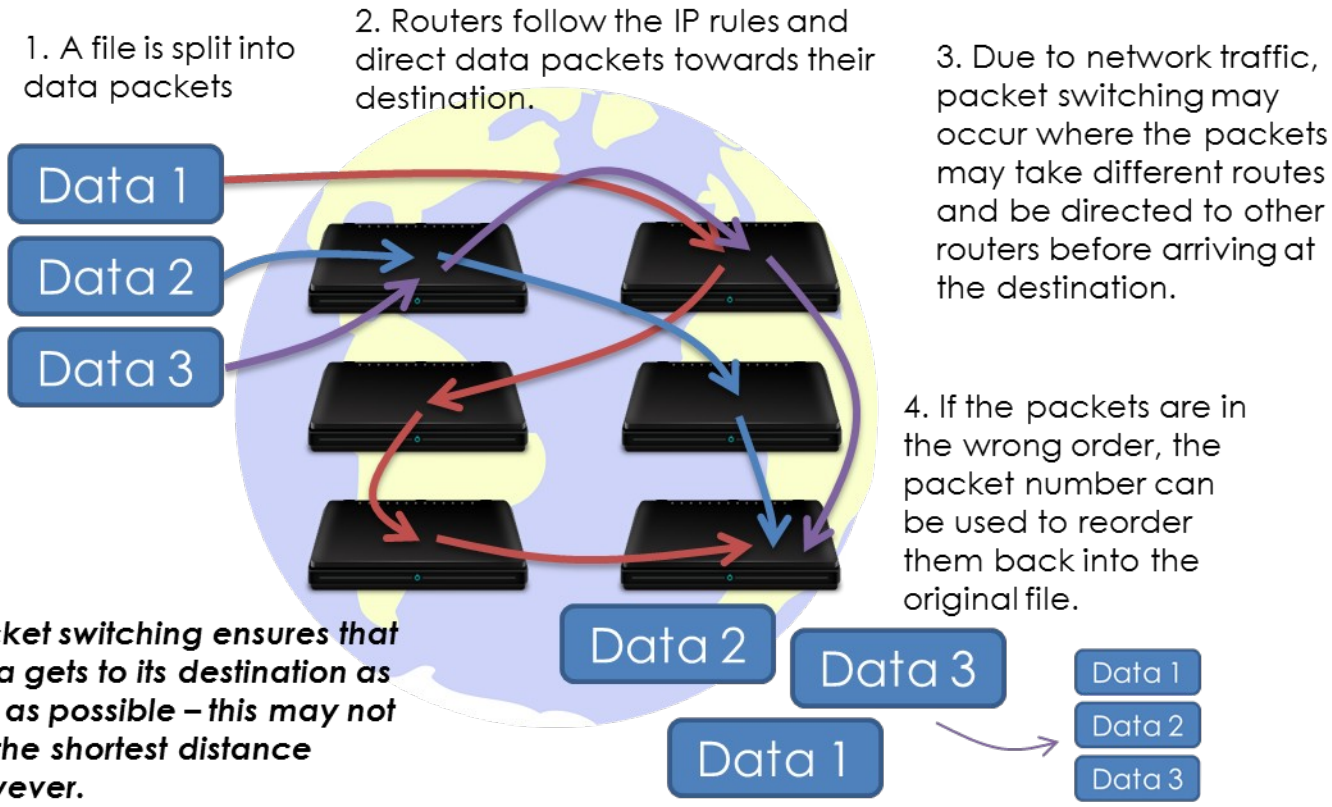
This rule is in place to ensure that data packets are directed towards their destination in the most appropriate way (which may not be the shortest way).

This is called 'Packet Switching' and we will now look at this in more detail.

Packet Switching

TCP/IP – Transmission Control Protocol / Internet Protocol

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World Wide Web (website) Protocols

HTTP – Hyper Text Transfer Protocol

This is the protocol which governs the way in which websites and web servers are accessed by web browsers. By conforming to the same standards it means that all browsers can access all websites on the WWW.

HTTPS – Hyper Text Transfer Protocol (SECURE)

This protocol is almost identical to HTTP. The only difference is that it ensures all data being sent is encrypted and so makes web data communication far more secure.

File Transfer (Upload, Download, Read and Write) Protocols

FTP – File Transfer Protocol

If you have your documents / files stored on a file server, the way you access them will be governed by the File Transfer Protocol. This standard (FTP) ensure that all clients (computers) on a network can access files from a server in the same way.

Email Protocols

POP – Post Office Protocol

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This is the protocol which governs how emails are accessed from a server. Email clients will use this protocol to access emails from a server in the same way.

IMAP – Internet Message Access Protocol

This is similar to POP but with one important difference. With IMAP, the email will remain on the server once it has been downloaded by a mail client. This allows other devices to also access the emails which is important for users with multiple devices – they can synchronise their emails across all devices.

SMTP – Simple Mail Transfer Protocol

This protocol governs how data is sent between mail servers. By conforming to the same standards, emails can be sent to and from all mail servers across the internet.

The Layers of a Network

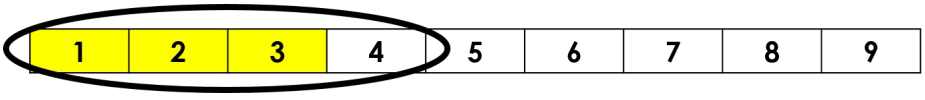
All of these protocols which enable the internet to function correctly can be divided into 'Layers of the Internet'. Each layer has similar functions and each layer provides a base for the layer above it – in other words, each layer does all the hard work for the layer above it!

Layer	Example Protocols	Layer Purpose
Application Layer – Layer 4	FTP, HTTP, SMTP	Provides user access to applications websites, files, email.
Transport Layer – Layer 3	TCP	Provides transport of data between devices by splitting files into data packets and checking that they have been sent and received correctly.
Network Layer – Layer 2	IP	Provides the routing of data across the network by making use of addressing.
Data Link Layer & Physical Layer – Layer 1	Ethernet	Provides the physical transport of data through electrical signals etc.

For example, if you are accessing a website (HTTP - layer 4), it requires data to be split into packets and sent across the internet (TCP – layer 3 / IP – layer 2), which in turns requires the physical transport of data via cables (cables / Wi-Fi – layer 1).

Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3

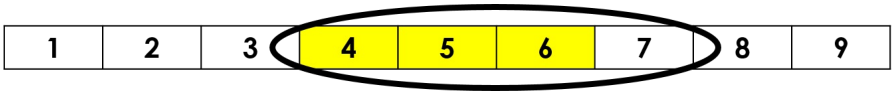


1. What is meant by the word 'Protocol' and why do networks need protocols? [2]

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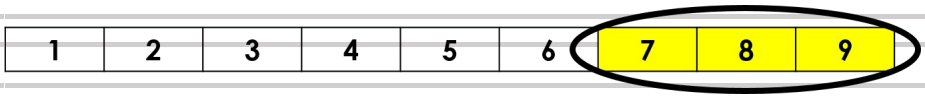
2. What is a data packet and what does a data packet consist of? [6]
3. Describe the process involved in packet switching. [4]

Question Zone 4-6



1. Describe the process involved in packet switching. [4]
2. Explain the need for channels within Wi-Fi frequency bands. [2]
3. Explain the reason for data packets and describe fully the various components found in a data packet, explaining why they are needed. [7]

Question Zone 7-9



1. Explain the reason for data packets and describe fully the various components found in a data packet, explaining why they are needed. [7]
2. Explain the various layers of a network. [4]
3. Discuss the issues involved in using wireless technology as a data transfer medium. [6]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples:** Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Protocols: *The rules that govern the workings of a network.*
Packet Switching: The method of routing data packets towards their destination according to the available routes at that time.
Data Transfer Medium: *The mechanism used to transfer data (via wires or Wi-Fi)*

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Stick answer sheet here

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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